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justice

Justice. To be ever ready to admit that another person is something quite different from what we read when he is there (or when we think about him). Or rather, to read in him that he is certainly something different, perhaps something completely different from what we read in him. Every being cries out silently to be read differently.
Simone Weil

My favorite Chinese restaurant serves authentic cuisine just like my parents used to make. When you order an entrée, they bring you a little appetizer and a dessert as well! It's a bargain, so I don't complain that the appetizer (crunchy noodles) and dessert (Jell-O) are not themselves authentic.

But one day I went there with a Chinese-speaking friend. When the appetizer came, it was not crunchies but delectable pickled cucumbers. Yet my friend had made no special request. And when the dessert came it was red bean soup—my favorite when I was a kid! Why had I not gotten this before?

I began to see a pattern: when I came with non-Asian friends, I would get the crunchies and Jell-O. But when I came with Asian friends, I got the good stuff without even asking.

And then I noticed that my Chinese friends were also offered a completely different menu—a secret menu—with more-authentic dishes. I looked around the restaurant and beheld a bizarre sight: people side by side in the same space having very different experiences—non-Asians ordering from a standard menu and getting Jell-O, but Asians ordering from the secret menu and enjoying red bean soup.

"You won't like the stuff on that menu," I was told. Even though I am of Chinese descent, the waiters had assumed, because I speak perfect English, that I must not be interested in the authentic Chinese food.

Mathematical spaces—in the home and in the classroom—can be like this restaurant. Whom do we allow a peek at the secret mathematical menu? With whom do we share mathematical delights—puzzles, games, toys? Whom do we let into our information circle about mathematics—news, videos, social media posts? Whom do we shepherd toward doing more mathematics, and whom do we discourage? What conscious or unconscious assumptions are we making?

Akemi was a student of mine who did math research with me as an undergraduate. Her innovative paper linking game theory (the mathematical modeling of decision making) and phylogenetics (the study of relationships between organisms) was published in a highly regarded mathematical biology journal. She went to a top research university to pursue a PhD in mathematics, so I was surprised when I learned that Akemi had quit after one year.

She told me that she had had many negative experiences. Her advisor was never willing to meet with her, and she had faced uncomfortable experiences as a woman. She told me one example:

> At the beginning of the course, I consistently got 10/10 on my homework assignments, which were all graded by the TA. One day, Jeff [a mutual friend] told me that he was hanging out with our TA and someone asked the TA how the analysis class was doing. He went on and on about some "guy" named Akemi and how perfect "his" homeworks were and how clearly they were written, etc. Jeff told him I was a girl and the TA was shocked. (Jeff told me this story because he thought it was funny that someone both didn't know my sex from my name and reacted so dramatically to finding out.) After that, I never got remotely close to 10/10 on my assignments and my exams were equally harsh—most of the reasons for docked points were vague, with comments like "give more detail." I didn't feel like my understanding of the material diminished that quickly or dramatically, but I suppose it's possible that happened and I'm just misinterpreting the situation.

If you feel that something is not right with this picture, you are experiencing a telltale sign of flourishing: a desire for justice. Justice is a basic human desire.

Justice means treating people equitably—giving each person what they are due. It means speaking up for the powerless, who are most likely to be treated unjustly. Some religious traditions, including mine, advocate caring for orphans, widows, immigrants, and the poor. I see these categories in the mathematical community as well: those without an advocate, those without a vibrant mathematical family, those who are new to mathematics, and those without resources or access to mathematics. These are the powerless in mathematics.

Some speak of justice in two categories: *primary justice* and *rectifying justice*.[1] Both are important. Primary justice involves right relationships: treating each person with dignity and care, and establishing social practices and institutions that support these aspirations. We flourish when we treat others justly and when we are treated justly.

Rectifying justice is spotting something wrong and trying to make it right. If primary justice were commonplace, rectifying justice would not be needed. But injustices are everywhere. Coercive power relationships are destructive. People and institutions may subtly promote inequities without realizing it.

Simone Weil realized that correcting injustice must involve changing how we view others. "Every being cries out silently to be read"—to be judged—"differently." Akemi wished to be read fairly by her TA, but the TA may not even have realized what he was doing. This is the problem of *implicit bias:* unconscious stereotypes that subtly affect our decisions. Before we criticize Akemi's TA, we have to realize that the problem of reading others differently begins with ourselves. I've taken diagnostic exams that revealed my implicit bias; these helped me see in a compelling way how I am biased even though I try not to be.[2]

We all have implicit biases. Numerous experiments confirm results of the following kind: when given two nearly identical resumes except that one has a positively stereotyped name and one has a negatively stereotyped name (that depends on context, but often female or minority), judges will rate the positively stereotyped resume higher. This happens even if the judges come from the negatively stereotyped group. Similar studies confirm that math achievement is correlated with teacher and parent stereotypes. For instance, a 2018 study of elementary school math test performance showed that teachers scored girls lower (and boys higher) compared to outside evaluators who didn't know the identities of the students, and the long-term effects of this bias persisted downstream: those girls were less likely to sign up for advanced math courses in high school.[3] A 2019 study showed that the gender gap in math performance increases substantially when middle-school students are assigned to teachers with stronger gender stereotypes, leading girls to develop lower self-confidence, underperform, and self-select into less demanding high schools.[4] Parental attitudes and stereotypes compound the problem. These issues disproportionately affect women and minorities.

If you believe that mathematics is for human flourishing, you will be dismayed, if you look at the demographics of those in mathematical studies or careers, that we aren't helping all people flourish. Across all races and ethnicities, college students intend to major in STEM (science, technology, engineering, mathematics) at the same rates, but rates of STEM-degree completion for underrepresented minorities are little more than half those for other groups.[5] Low-income and first-generation college students complete college degrees at far lower rates overall than non-first-generation students, a challenge that also carries over to STEM disciplines.[6] Women drop out of PhD programs in STEM at higher rates than men. At the end of the STEM pipeline, the demographics are overwhelmingly white and male from higher socioeconomic backgrounds.[7] We are losing many with an interest in STEM, but the losses are more pronounced in marginalized groups—a deeply inequitable situation.

Even to talk percentages makes it sound like we live in a zero-sum world, as if one person going into a STEM profession means another person does not. Contrary to this perception, the world will need drastically more people with mathematical skills as we become ever more STEM reliant. The 2012 report *Engage to Excel* from the President's Council of Advisors on Science and Technology estimates that in order to maintain its preeminence in STEM, the US alone will need to produce an *additional* million STEM graduates over the next decade than it is currently expected to produce. That's not even emphasizing the obvious: we all lose when talent is overlooked, when we don't nurture those whose discoveries could benefit us all. A society cannot flourish if its people are not living up to their potential.

To rectify injustices, we have to talk about hard things that might separate us, like race, gender, sexual orientation, social class, the rural-urban divide, and the related ways in which some are marginalized in mathematics. Such conversations can bring up complicated emotions. We have to become more comfortable with talking about difficult topics, listening to one another's experiences, and recognizing the pain that's there. If you want to treat others with dignity and they're hurting, you don't ignore their pain—you ask: "What are you going through?"

It's not enough to say, "I don't think about that stuff—I treat everyone the same," because in any community, how any one member is doing affects the whole. Those of us in marginalized groups don't have the luxury of saying, "I don't think about that stuff," because that stuff affects us on a daily basis.

So let me encourage all of us to try having these conversations, to be quick to listen, slow to speak, and quick to forgive each other when we say something stupid. That'll happen if you start to have conversations, and we just have to show grace to one another if we make mistakes—it's better than not talking.

I'll share some of my experiences around race. I'm Chinese American. I grew up in a white and Latino part of Texas, and

I realized early on that my family had different customs than my friends—my clothes were different, the food in my lunch box was different—and that these things were isolating me. I was getting picked on all the time. I wanted to be white. I had few role models for being Asian American. I remember that my mother would always call me to the TV when the journalist Connie Chung was on, because it was so rare to see anyone who looked like us in the media. My dad would clip newspaper articles when any Asian Americans made the news. I was embarrassed to be Asian, so I tried hard to act white, even if I couldn't look white. This meant I would publicly deny anything Asian—not showing interest in Chinese food, not talking about Chinese customs—while I modulated my hair, clothes, and idioms of speech to be like those of my friends.

But in Chinese communities, I also don't fit in. I don't speak Chinese. I don't act Chinese. At Chinese restaurants, I'm viewed as white. That's why at authentic Chinese restaurants, I never get the secret menu. Asian Americans like me often feel like we live between two cultures, always viewed as outsiders, never fully fitting into either.

There are ways in which I benefited from being Asian. People assume I'm good at math because I'm of Asian descent. I never had anyone discourage me from taking a math class (as has happened to women friends of mine) or question whether I belonged at a math conference (as has happened to some of my African American friends). But even as I've benefited, I know that some of my Asian friends feel embarrassed when they don't live up to this stereotype.

The first time I didn't feel like a minority was when I moved to California, where there are so many Asian Americans. In Texas I would commonly get the well-meaning question "Your English is so good! Where are you from?" I'd say, "Texas," and inevitably, I'd hear, "No, I mean, where are you *really* from?" That happens less in California, and there's a feeling of freedom I have in not having to counter those verbal stings.

These days I'm used to being at math conferences and seeing a sea of white faces. So even I was a bit surprised that when I was elected as president of the Mathematical Association of America (MAA), a prominent blogger on race issues for Asian Americans, who goes by the alias Angry Asian Man, wrote a post about it.

Angry Asian Man looked at the photos of 100 years of past presidents on the MAA website, and given how many Asians he expected to find in math, he was astonished to note that they were all white except for me, and wrote a blog post sarcastically titled "Finally, an Asian Guy Who's Good at Math."[8]

I am the first president of color of the MAA. Minorities, including Asians, are easy to overlook when you think about who would make a good leader. This may not be intentional, but when we are asked to think about who is fit for this or that role, we often think of people just like those who have already been in office, and implicit bias creeps in. We don't realize what we might gain by having diverse people, new expertise, fresh ideas to draw from. The field of mathematics is itself poorer because of the voices that are not present. The math education professor Rochelle Gutiérrez reminds us that math needs a diversity of people in order to grow in new ways, not just that people need math: "The assumption is that certain people will gain from having mathematics in their lives, as opposed to the field of mathematics will gain from having these people in its field."[9]

I raise this discussion out of deep affection for all communities of math explorers. I want us to flourish, to make space to welcome new explorers from all different backgrounds, and we can do better.

Besides falling prey to implicit bias, there are other ways that mathematical communities misjudge people.

We assume that grades are a measure of mathematical promise. This is not a correct assumption, for many reasons. I used to worry that a student getting Bs in college math courses wouldn't be successful in graduate school. But I've seen many now who have gotten their PhDs and have flourished as mathematicians.

Grades are a measure of progress, but not a measure of promise. Everyone is in a different place in their mathematical knowledge. You see the snapshot, but you don't see the trajectory. You can't know how people will flourish in the future. But you can help them get where they want to go. When someone has trouble in mathematics, we should bolster our support, not lower our expectations.

It is easy to make assumptions about why people aren't performing mathematically, based on our own experiential vantage point. We can't imagine alternatives that we haven't experienced. We can't always know what personal issues someone is facing. A student once tearfully told me about the hours she spent filling out financial aid forms because her parents didn't speak English. Her immigrant family expected her to spend every weekend at home, and her home wasn't an environment conducive to homework. College was a culture shock, with many unwritten rules. This student was navigating many complex realities. Her performance wasn't the best she could offer, because of those realities.

Christopher Jackson has his own complex reality. Studying math in prison, he's mostly isolated from others. He hasn't been in school for a decade or more. He's developing his own way of learning and expressing mathematics. Expecting him to communicate math in the same way that I'm used to is unrealistic.

I'm sure that traditional assessments would not give a real picture of what he knows.

Low performance should not be used as an excuse to rob students of opportunity. The practice of tracking—sorting students into a dead-end course progression based on low performance—happens in some K–12 schools and is highly inequitable. Bias creeps into who is assigned to the "low" track. Those students are put into dead-end courses that do not prepare them for college and careers, are given access to less experienced teachers, and are assigned rote memorization exercises rather than rich meaning-making, sense-making activities. They cannot flourish mathematically. Tracking is a coercive practice that must be stopped.[10]

We assume that learning mathematics doesn't involve culture. This is a common assumption, especially if you are not part of a marginalized group. It leads to inaccurate assessments of student knowledge. A mathematician friend shared this example with me:

> On an examination, I asked the classical Fermi problem "Estimate how many piano tuners live in this city." A student timidly raised his hand. He whispered to me, "Is a piano tuner a device or a person?" Other students thought good piano players would tune their own pianos, like guitar players do. Some students thought piano tuners would work in a music store. Few students had a sense of how often a piano might need tuning, or how long it would take to tune a piano. This example opened my eyes to how important background experience can be in dealing with questions that may appear to be mathematical, but instead bring up all sorts of cultural or experiential issues.

I could have been one of those perplexed students, since a piano wasn't a household item for me. Now imagine a student without the requisite cultural experiences who constantly encounters obstacles like these. Would they feel like they belong? Cultural barriers are impossible to avoid, but if we are aware of them, we can mitigate their effects.

The math education professor William Tate points out that such experiences are common to African American children in mathematical spaces, who often encounter instruction based on white middle-class norms, and he contends that connecting pedagogy to the lived realities of African American students is essential for equitable instruction.[11] He advocates that teachers take a "centric" perspective: allowing and expecting students to center their problem solving in terms of their own cultural and community experiences, and encouraging students to think about how the same problem might be viewed from the perspectives of different members of the class, school, and society. For example, a teacher could reframe the problem about piano tuners as an estimation problem whose subject the students choose as relevant to their daily lives or struggles.

We assume that certain people won't be successful in mathematics, and we push them away from math. "You won't like the stuff on that menu." But if you believe that mathematics is for human flourishing, why would you do that?

In 2015, I had the great pleasure of running MSRI-UP (Mathematical Sciences Research Institute—Undergraduate Program), a summer research program for students from underrepresented backgrounds: Hispanic, African American, and first-generation college students. Later, I asked them to tell me about obstacles they'd faced in doing mathematics. One of them, who did wonderful work that summer, told me about her experience in an analysis course after she returned to her university:

> Even though the class was really hard, it was more difficult to receive the humiliations of the professor. He made us feel that we were not good enough to study math, and he even told us to change to another, "easier" profession.

As a result of this and other experiences, she switched her major to engineering.

Let me be clear: there is no good reason to tell someone that she shouldn't be doing mathematics. That's her decision—not yours. You may not know what she's capable of. One of my friends who is now a math professor described this incident that happened to him when he was a student:

> This faculty member had one of those private in-the-office conversations with me that begins with "I think it may be only a kindness to tell you that . . . ," followed by a stated concern that I was not really cut out for a career in mathematics. I've not done all that badly since then, and in fairness I have to add that the faculty member sought me out in later years to apologize for the comment. I consider the person a friend, but when I'm working with our graduate student training program, I do stress that any conversation that begins with "I think it only a kindness to tell you that" will almost never be a kindness.

Look at my friend now—a successful mathematician. It's too easy for such pronouncements to reflect personal biases and limited information.

Oscar, another student from MSRI-UP, told me about his experience as a math major who, unlike his peers and because of his background, did not enter college with any advanced placement credit:

> I noticed how different my trajectory was, however, while I was in a Complex Analysis course. A student was presenting a solution on the board which required a bit of a complicated derivation halfway through. They skipped over a number of steps, saying, “I don’t think I need to go through the algebra . . . we all tested out of Calculus here anyway!,” with my professor nodding in agreement and some students laughing. I quietly commented that Calculus was my first course here. My professor was genuinely surprised and said, “Wow, I did not know that! That’s interesting.” I was not sure whether to feel proud or embarrassed by the fact that I was not the “typical math student” that was successful from the beginning of their mathematical career. I felt a sense of pride in knowing that I was pursuing a math degree despite my starting point, but I could not help but feel as though I did not belong in that classroom to begin with.

The reason Oscar was in that class to begin with was the active support of another professor. Oscar said:

> She presented me with my first research opportunity and always encouraged me to study higher math. I was also able to confide in her about a lot of the internal struggles I had with being a minority in mathematics since, as a female, she had a similar experience herself! My complex analysis professor became one of my mentors as well. I think it was just an interesting moment because she didn’t realize how her reaction to the situation could have hurt me (and I don’t think she’s necessarily at fault!). It was more that

> her reaction piled onto the insecurities I held in regards to being a minority with a weak background in math.

Actually, Oscar didn't have a "weak" background—he had a standard background. I'm pleased to say that Oscar and his team from that summer have published a paper on their research, and he is now in graduate school.

You hear in Oscar's story the importance of having an advocate, someone who says, "I see you, and I think you can flourish in mathematics." Everyone can use this encouragement, but this can be especially important for marginalized groups who already have so many voices telling them they don't belong. Can you be that advocate?

We must be mindful to not set up structures for learning that disadvantage people with weaker backgrounds or make them feel out of place. When I was teaching at Harvard, there was a regular calculus class, an honors calculus class called Math 25, and on top of that—for those with very strong backgrounds—a super honors class called Math 55. Ironically, I regularly encountered students *in the honors track* who felt that they didn't belong in the math major, because they hadn't placed into the super honors track. I had to keep reassuring them that "background is not the same as ability." I wish that college and grad school admissions would remember this too. The mathematician Bill Velez says this about barriers at the graduate level: "In mathematics we value creativity, yet we evaluate students on knowledge. Departments erect barriers to keep down applications, and it works. Top-rated departments have few minority students."

Seeking justice can be a motivation to study mathematics, to rectify the injustices that exist in spaces of mathematical learning for the powerless in mathematics: the "orphans" who need advocates, the “widows” who need mathematical community, the “immigrants” new to mathematics, and the “poor” who face barriers of opportunity. Those who pursue justice in mathematics cultivate the virtues of *empathy for the marginalized* and *concern for the oppressed.* We sometimes don’t recognize the constant burden of oppression that is experienced by the powerless until we start opening our eyes to see what they see. Those of us with power need to assist those who have no power.

The math teacher Josh Wilkerson engages his AP Statistics classes in a service-learning project, partnering with a homeless ministry in Austin, Texas, for survey research and data analysis. The students do a lot of “nonmath” reading on homelessness, to break down their assumptions on why people tend to become homeless. They also administer a survey and speak with a formerly homeless person face-to-face. As Josh puts it, “Hopefully, they recognize that behind every data point is a person, that person has a story, and that story is important.”

Seeking justice builds in us a *willingness to challenge the status quo.* Many injustices are deeply embedded in the way institutions operate, whether that be school or workplace or home. People get different menus all the time, and no one says a thing. Long-standing inequities are hard to recognize because we operate within them and they’re what we’ve always known. We need voices crying in the wilderness to call attention to the ways that we must change in order to treat each person with dignity and care in mathematical spaces.

I dream of that day when the secret menu will no longer be secret—when all people will be encouraged to develop their mathematical tastes so that someday they can be connoisseurs, even chefs, of mathematical cuisine.

RENTAL HARMONY

We've talked about just behavior in mathematical communities, but you can also use mathematics to study notions of justice. There's an area at the intersection of mathematics and economics called "fair division," which is concerned with how to divide things equitably among several people. A prototypical question is "How do you cut a cake fairly?" Math is involved in modeling people's preferences with sets or functions. I began doing research in this area when I encountered this problem:

> You and your college friends decide to rent a house together, and you have found a candidate house. However, the house has rooms of different sizes and with different features, and each of you has different preferences. Is it always possible to split the rent and price the rooms in such a way that each person will want a different room?

The answer is yes, under mild conditions—here's a result I proved in 1999:

> *Rental Harmony Theorem*
>
> Suppose the following conditions hold:
>
> 1. (Good house) In any rent division, each person finds some room acceptable at the proposed rent.
> 2. (Closed preferences) If a person prefers a certain room even as the rents change and approach a limiting rent division, then that person will continue to prefer that room in the limiting rent division.
> 3. (Miserly tenants) A person will always prefer a free room to a nonfree room.
>
> Then there exists a rent division in which each person will prefer a different room.

The proof involves ideas from geometry and combinatorics (the study of ways of counting things), and it yields a procedure for finding a fair division of rent. When a *New York Times* reporter used my procedure to solve his rent division problem, he wrote about it in an article and published an interactive app that implements this procedure. I encourage you to try the web app.[a]

By the way, if you drop the miserly tenants condition, the theorem is still true, but you'll need to allow negative rents; in other words, you can still find a solution, but you may need to pay someone to live with you!

a. The rental harmony result may be found in Francis E. Su, "Rental Harmony: Sperner's Lemma in Fair Division," *American Mathematical Monthly* 106 (1999): 930–42. The *New York Times* article is Albert Sun, "To Divide the Rent, Start with a Triangle," *New York Times,* April 28, 2014, https://www.nytimes.com/2014/04/29/science/to-divide-the-rent-start-with-a-triangle.html; the interactive web app can be found at https://www.nytimes.com/interactive/2014/science/rent-division-calculator.html.

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I appreciate your efforts in trying to call up here for me; what everybody's doing on the outside is really starting to have an effect and I should be out soon. I got your letter on 8/31 (they had me in medical observation from 8/16 to 8/28 and I wasn't receiving any of my mail then). Hopefully, I'll be able to send you this in an e-mail (I've been on a hunger strike 24 days, and now I've seen my unit manager 3 times in the last 5 days—before then I hadn't seen him in a month and half) because it seems me and the administration have reached a resolution and this should be over very soon. I'm supposedly going to find out tomorrow.

One of the important things I'm learning from you is that mathematics is around and about ideas. Not necessarily just the ideas inside of mathematics but also the ideas parallel to mathematics. (I'm about to digress.) I remember my first incursion into the realm of ideas. . . . I was 16 and I was in a group home; my caseworker at the time gave me 3 books, [including] *The Art of War* by Sun Tzu, great book, even though the title might throw some people off, on a philosophical level or even a metaphysical level for dealing with strife, whether it be internal, external, personal, interpersonal or whatever. . . . These books ignited my interest in philosophy, which then went on to politics, economics, business, and then eventually back to mathematics.

(9/9/18: I came off hunger strike 9/7/18. I'm designated to another institution—I should be on a bus leaving out of here this coming week. I started this letter while I was still not eating, because I had an idea of what I wanted to write, but it was getting difficult because my glucose levels were averaging around 65.)

And through every new idea that I've encountered, whether they be ideas within a discipline or ideas across disciplines, I started to notice principal ideas: order, relation, organization, structure, process. . . .

And from what I'm gleaning from you about mathematics being around and about ideas, I have to ask, do you think mathematics unites these ideas?

Chris